

**FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

**MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS
(Multiple Choice Questions for SDE Candidates)**

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS

(Multiple Choice Questions for SDE Candidates)

1. The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$ is :

- (A) 0. (B) 1.
(C) 2. (D) 3.

2. The rank of the matrix $\begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & 2 & 0 \end{bmatrix}$ is :

- (A) 3. (B) 4×3 .
(C) 2. (D) 1.

3. Rank of the matrix $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ -1 & -3 & -4 & -3 \end{bmatrix}$ is :

- (A) 1. (B) 2.
(C) 3. (D) 4.

4. The points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ are collinear if and only if the rank of the matrix $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$ is :

- (A) < 3 . (B) ≤ 3 .
(C) > 3 . (D) ≥ 3 .

5. If a matrix A has a non-zero minor of order r , then :

- (A) $\rho(A) = r$. (B) $\rho(A) \geq r$.
(C) $\rho(A) < r$. (D) $\rho(A) \leq r$.

6. Which of the following is false :

(A) $\rho(A+B) \leq \rho(A) + \rho(B)$.

(B) $\rho(A') = \rho(A)$.

(C) $\rho(A+B) = \rho(A) + \rho(B) - 4$, if A and B are matrices of rank 4.

(D) $\rho(A-B) \leq \rho(A)\rho(B)$.

7. Rank (AA') = _____.

(A) Rank A.

(B) Rank A' .

(C) 1.

(D) None.

8. Rank (AA^0) = _____.

(A) Rank A^0 .

(B) Rank A.

(C) Rank A' .

(D) None.

9. The system $AX = 0$ in n unknowns has a non-trivial solution if :

(A) $\rho(A) > n$.

(B) $\rho(A) = n$.

(C) $\rho(A) < n$.

(D) None of these.

10. A system of M homogeneous linear equations $AX = 0$ in n unknown has only trivial solution if :

(A) $m = n$.

(B) $m \neq n$.

(C) $\rho(A) = m$.

(D) $\rho(A) = n$.

11. The system of equations $x + 2y + z = 9$ can be expressed as :

$$2x + y + 3z = 7$$

(A) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$.

(B) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$.

(C) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$.

(D) None.

12. If A is a square matrix of order n and λ is a scalar, then the characteristic polynomial of A is obtained by expanding the determinant.
- (A) $|\lambda A|$. (B) $\lambda |A|$.
 (C) $|\lambda A - I_n|$. (D) $|A - \lambda I_n|$.
13. The characteristic roots of Skew-Hermitian matrix are either :
- (A) Real or zero. (B) Real or non-zero.
 (C) Pure imaginary or zero. (D) Pure imaginary non-zero.
14. The matrix $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ satisfies the equation :
- (A) $A^2 + 5A + 7I = 0$. (B) $A^2 + 5A - 7I = 0$.
 (C) $A^2 - 5A - 7I = 0$. (D) $A^2 - 5A + 7I = 0$.
15. The product of all the characteristic roots of a square matrix A is equal to :
- (A) 0. (B) 1.
 (C) $|A|$. (D) $\frac{1}{|A|}$.
16. If eigen value of matrix A is λ , then eigen value of $P^{-1}AP$ is :
- (A) 1. (B) λ .
 (C) $\frac{1}{\lambda}$. (D) 0.
17. A polynomial equation in x of degree n always have :
- (A) n distinct roots. (B) n real roots.
 (C) n complex roots. (D) None.
18. A zero of the polynomial $x^3 + 2x - i$ equals :
- (A) $-i$. (B) 1.
 (C) $1 - i$. (D) None.
19. If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$ then $\alpha\beta + \beta\gamma + \gamma\alpha$ equals :
- (A) $\frac{-p}{q}$. (B) $-p$.
 (C) q . (D) $-q$.
20. A polynomial equation whose roots are 3 times those of the equation $2x^3 - 5x^2 + 7 = 0$ is :
- (A) $3x^3 - 15x^2 + 21 = 0$. (B) $2x^3 - 15x^2 + 189 = 0$.
 (C) $2x^3 + 15x^2 - 189 = 0$. (D) None.

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2021

Mathematics

MAT 4B 04—THEORY OF EQUATION, MATRICES AND VECTOR CALCULUS

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type)

*Answer all the **twelve** questions.*

Each question carries 1 mark.

1. If α, β, γ are the roots of $2x^3 + 3x^2 - x - 1 = 0$. Find the equation whose roots are $\alpha^2, \beta^2, \gamma^2$.
2. State Descarte's rule of signs.
3. Find a cubic equation, two of whose roots are given by $1, 3 + 2i$.
4. What do you mean by reciprocal equation of first type ? Give example.
5. What is the rank of the identity matrix of order 101 ?
6. If $A = [a_{i,j}]$ is an $m \times n$ matrix and $a_{i,j} = 7$ for all i, j then rank of A is _____.
7. A system of m homogeneous linear equations in n unknowns has only trivial solution if _____.
8. For what values of a the system of equations $ax + y = 1, x + 2y = 3, 2x + 3y = 5$ are consistent.
9. If the number of variables in a non-homogeneous system $AX = B$ is n then the system possesses a unique solution if _____.
10. Find the parametric equation of a line through $P(1, 1, 1)$ and parallel to the z -axis.
11. Find the unit tangent vector of the helix $r(t) = (\cos t + t \sin t)i + (\sin t - t \cos t)j + tk, t > 0$.
12. Write equations relating spherical and cylindrical co-ordinates.

(12 × 1 = 12 marks)

Turn over

Part B (Short Answer Type)*Answer any nine questions.**Each question carries 2 marks.*

13. Solve $6x^3 - 11x^2 - 3x + 2 = 0$. Given that the roots are in harmonic progression.
14. Find the equation whose roots are the roots of $x^3 + 3x^2 - 2x - 4 = 0$ increased by 5.
15. If α, β, γ are the roots of $x^3 + qx + r = 0$. Find the equation whose roots are $(\beta - \alpha)^2, (\gamma - \alpha)^2, (\alpha - \beta)^2$.
16. If $A = \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}$. Find A^{-1} .
17. Prove that the characteristic roots of Hermitian matrix are real.
18. If α is an eigen value of a non-singular matrix A , prove that $\frac{|A|}{\alpha}$ is an eigen value of $\text{adj } A$.
19. Show that the product of characteristic roots of a square matrix of order n is equal to the determinant of the matrix.
20. Find the value of a for which $r(A) = 3$ where $A = \begin{pmatrix} 2 & 4 & 4 \\ 3 & 1 & 2 \\ 1 & 0 & a \end{pmatrix}$.
21. Find the velocity and acceleration vectors of $r(t) = (t+1)i + (t^2 - 1)j + 2tk$ at $t = 1$.
22. Find the rectangular co-ordinates of the centre of the sphere $r^2 + z^2 = 4r \cos \theta + 6r \sin \theta + 2z$.
23. Evaluate $\int_0^\pi (\cos ti + j - 2tk) dt$.
24. Find the principal unit normal for the circular motion $r(t) = (\cos 2t)i + (\sin 2t)j$.

Part C (Short Essay)*Answer any **six** questions.**Each question carries 5 marks.*

25. If α, β, γ are the roots of $x^3 - x - 1 = 0$. Find the equation whose roots are $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$.

26. Solve the equation $x^2 - 12x - 65 = 0$ by Cardan's method.

27. Solve $x^3 + 6x^2 + 3x + 18 = 0$.

28. Prove that the rank of the transpose of a matrix is equal to the rank of the same matrix.

29. Find the rank of $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & -2 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{pmatrix}$.

30. Using matrix method solve :

$$2x - y + 3z = 9$$

$$x + y + z = 6$$

$$x - y + z = 2.$$

31. Find the point in which the line $x = 1 + 2t, y = 1 + 5t, z = 3t$ intersects the plane $x + y + z = 2$.

32. Find the distance from the point $S(2, 1, 3)$ to the line $x = 2 + 2t, y = -1 + 6t, z = 3$.

33. Find the eigen values and eigen vectors of $\begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$.

(6 × 5 = 30 marks)

Turn over

Part D

*Answer any **two** questions.*

Each question carries 10 marks.

34. Solve the equation $6x^5 - 41x^4 + 97x^3 - 97x^2 + 41x - 6 = 0$.

35. Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 2 & 3 & 5 \\ 3 & 1 & 2 \\ -1 & 2 & 2 \end{pmatrix}$ and hence evaluate A^{-1} .

36. Find the binormal vector and torsion for the space curve $r(t) = \left(\frac{t^3}{3}\right)i + \left(\frac{t^2}{2}\right)j$.

(2 × 10 = 20 marks)